# Experiment No-21 : Towers of Hanoi

## Aim:

To write a Prolog program to implement the Towers of Hanoi problem.

## Programming Code:

```
% hanoi(N): move N disks from the left peg to the right peg using the center peg
hanoi(N) :- move(N, left, right, center).

% move(N, From, To, Via)
move(0, _, _, _) :- !.
move(N, From, To, Via) :-
    M is N - 1,
    move(M, From, Via, To),
    format("Move disk ~w from ~w to ~w~n", [N, From, To]),
    move(M, Via, To, From).

% Minimum number of moves required for N disks = 2^N - 1
count_moves(N, Moves) :- Moves is (2 ** N) - 1.
```

**OUTPUT:**

```
$ swipl -q exp21.pl

?- hanoi(2).
Move disk 1 from left to center
Move disk 2 from left to right
Move disk 1 from center to right
true.

?- hanoi(3).
Move disk 1 from left to right
Move disk 2 from left to center
Move disk 1 from right to center
Move disk 3 from left to right
Move disk 1 from center to left
Move disk 2 from center to right
Move disk 1 from left to right
true.

?- count_moves(3, M).
M = 7.

?- count_moves(5, M).
M = 31.

?- count_moves(10, M).
M = 1023.

$
```

## Result

The Towers of Hanoi problem was implemented successfully using Prolog. The recursive program prints every move needed to shift all the disks from the left peg to the right peg, and the number of moves for N disks is 2^N - 1.